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Optimal Transport PINNs: A Data-Efficient Approach to High-Fidelity Turbulence Modeling

With the rapid expansion of data-driven fluid mechanics, turbulence modelling remains a fundamental challenge in computational fluid dynamics (CFD). Traditional methods like Reynolds-Averaged Navier-Stokes (RANS) and Large Eddy Simulation (LES) balance computational efficiency with reduced accuracy compared to Direct Numerical Simulation (DNS). Although DNS provides highly accurate turbulence data, its computational cost is impractical for many real-world applications. Physics-Informed Neural Networks (PINNs) integrate physical laws into deep learning models, offering a data-efficient alternative. However, they often face training instability and convergence challenges, especially in high-Reynolds-number flows. This study proposes an Optimal Transport (OT)-based loss function to enhance PINNs' accuracy and stability in turbulence modelling, even with 5% noisy data. By leveraging OT theory, we improve gradient flow alignment and mitigate spectral bias, ensuring more robust training. Additionally, we integrate the Sparse Identification of Nonlinear Dynamical Systems (SINDy) framework to extract interpretable reduced-order models from PINN solutions. Our evaluation utilizes a 2D slice from a 3D DNS simulation of homogeneous isotropic turbulence on a 1024×1024 grid at $z = 0$, featuring periodic boundary conditions and velocity components u , v and w . We validate our approach on benchmark turbulence cases, including fully developed channel flow and decaying isotropic turbulence, comparing results against high-resolution DNS data. Findings indicate that OT-PINNs deliver improved accuracy in velocity statistics and energy spectra while significantly reducing computational costs. This study highlights the potential of integrating PINNs, OT, and SINDy as a scalable alternative to DNS for high-fidelity turbulence modelling, even in noisy conditions.